Course: DSAI501 - How To Use Big-Data and AI to Make Better Business Decisions (ML and Data Science) **Assignment: Final group assignment - Data Science Group Assignment (Option C: Identifying Patterns in Global E-Commerce Sales)** **Group Members: Prem Jasani, BMT28115; Radhika Gupta, BMT28063; Rishi, BMT28067, Lars Godoy, BMT28046**	**Date: 16/10/2026** **Professor: Hunter Yan**

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Identifying Patterns in Global E-Commerce Sales

Customer segmentation of a UK online retailer with RFM and K-Means

Executive Summary

This report segments the customers of a UK based online gift and homeware retailer to show where revenue comes from and what the business should do about it. Working from the Online Retail II dataset of 1,067,371 transactions (Chen, 2012), we removed duplicates, cancellations, non product line items and records that cannot be tied to a customer, reaching a reliable core of 776,577 transactions, 5,852 customers and £17.07 million in revenue. For each customer we engineered Recency, Frequency and Monetary (RFM) features and segmented them with K-Means. We chose four segments: the silhouette score actually favours a coarser two-way split, but two groups cannot drive four different actions, so we let the business question set the granularity. The picture is highly concentrated. One segment, the Champions, is roughly 20% of customers but 73% of revenue, while a third of the base is effectively dormant. Each segment points to a different action, from protecting Champions to winning back a valuable At-Risk group that holds about £2.9 million in historic revenue. A market basket analysis adds specific cross-sell pairs. The central caveat is that this is a revenue only, behavioural view of a single retailer, so the findings should guide experiments rather than stand as final answers.

1. Introduction

Online retail records every purchase, yet most of that data is never examined closely. When it is, it usually shows that a business depends on far fewer customers than it assumes. Acquiring a new customer typically costs more than keeping an existing one, so knowing which customers are valuable, which are slipping away and which were never worth much changes how a company spends its marketing and retention budget. The aim of this project is to turn two years of raw transactions into a small number of customer segments that a marketing team can actually act on.

We use the Online Retail II dataset from the UCI Machine Learning Repository (Chen, 2012), which contains all transactions for a UK based, mainly online retailer of giftware between 1 December 2009 and 9 December 2011. Each row is a product line on an invoice, with an invoice number, product code and description, quantity, unit price in pounds sterling, a customer identifier and the customer's country. The dataset is well suited to this task because it is transactional, large enough to be realistic at over one million rows, and messy in the ways real operational data is messy, which makes the cleaning itself a meaningful part of the work.

Our modelling follows the RFM framework, a long standing method in database marketing (Hughes, 1994) that the dataset's own creators applied to this retailer (Chen et al., 2012). We summarise each customer by how recently they bought, how often, and how much they spent, then group similar customers with K-Means clustering (MacQueen, 1967; Lloyd, 1982). The deliverable is not the clustering itself but the translation of those clusters into segment specific recommendations.

2. Data Description and Preprocessing

The raw data combines two worksheets, 2009 to 2010 and 2010 to 2011, for a total of 1,067,371 rows across eight fields. Profiling it surfaced the issues a retention team would have to resolve before trusting any number. Customer ID is missing on 243,007 rows (22.8%). Invoice numbers beginning with

the letter C mark cancellations and account for 19,494 rows. There are 22,950 rows with a quantity of zero or less and 6,207 with a price of zero or less, both of which represent returns, samples and manual adjustments rather than sales. The two yearly worksheets also overlap at the December 2010 boundary, sharing 1,088 invoice numbers, so a naive concatenation would double count those orders.

We cleaned the data in a documented sequence and recorded how many rows each rule removed, shown in Figure 1. Duplicates were removed first, de-duplicated on transaction content rather than on a naive row match so that the boundary overlap between the two sheets is caught, which removes 34,335 rows in total. Cancellations were removed next because the goal is to model purchasing, although return behaviour is quantified in the results. We then kept only genuine product codes, removed non positive quantities and prices, and finally removed rows with no Customer ID, since customer level segmentation requires an identifiable customer. Because cancellations are removed before the quantity and price rules, the later steps show only the residual non positive rows, not the raw totals cited above. The result is 776,577 clean transactions covering 5,852 customers, 36,594 orders and £17.07 million in revenue.

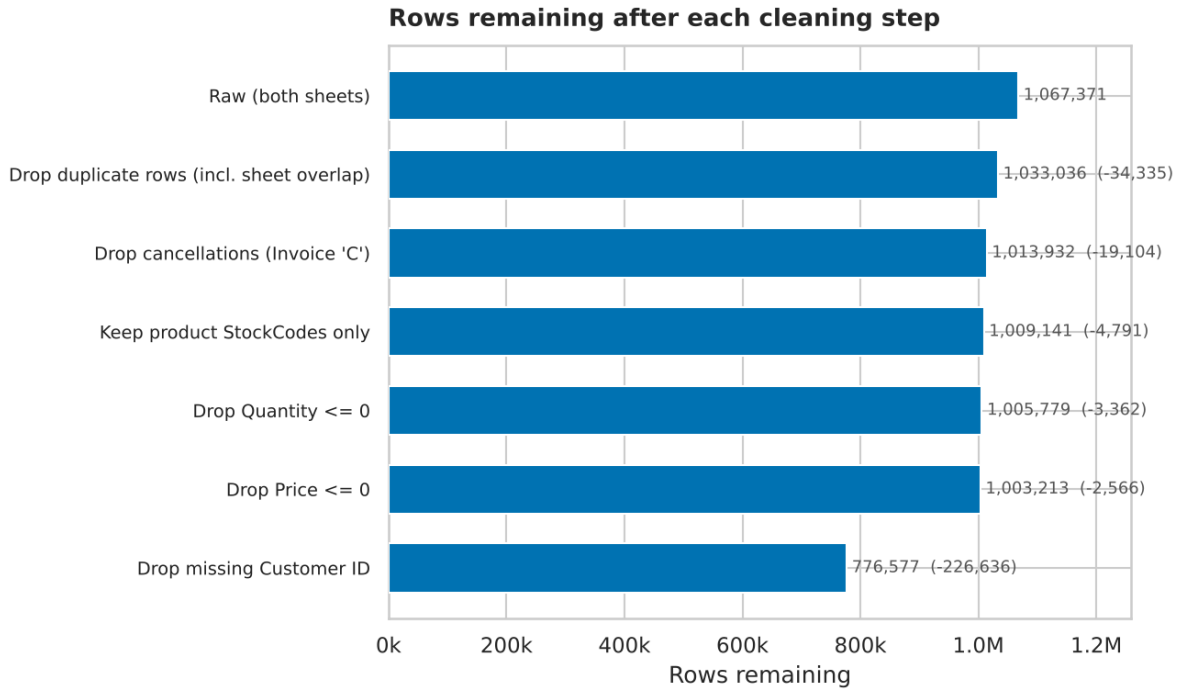


Figure 1. Rows remaining after each cleaning step.

One preprocessing decision deserves emphasis because it is also a limitation. The unidentifiable transactions are not negligible, but they are smaller than their row count suggests: they hold 13.3% of valid revenue despite being 22.8% of rows, which tells us that anonymous purchases skew toward small, one off baskets. We proceed with the identifiable customers while remembering that roughly an eighth of revenue sits outside the analysis.

3. Exploratory Data Analysis

Revenue is strongly seasonal and trends upward into the fourth quarter, peaking in November as retailers stock up before Christmas (Figure 2). The two partial months at the start and end of the window are visible as dips and should not be read as decline. Geographically the business is a UK company first and an exporter second: the United Kingdom alone accounts for 84% of revenue, and

the next largest markets, the Republic of Ireland, the Netherlands, Germany and France, trail far behind (Figure 3). These views use the cleaned, identified-customer transactions; including the unidentified but valid sales would raise revenue by about 13%, but those orders cannot be attributed to a customer or a clear pattern.

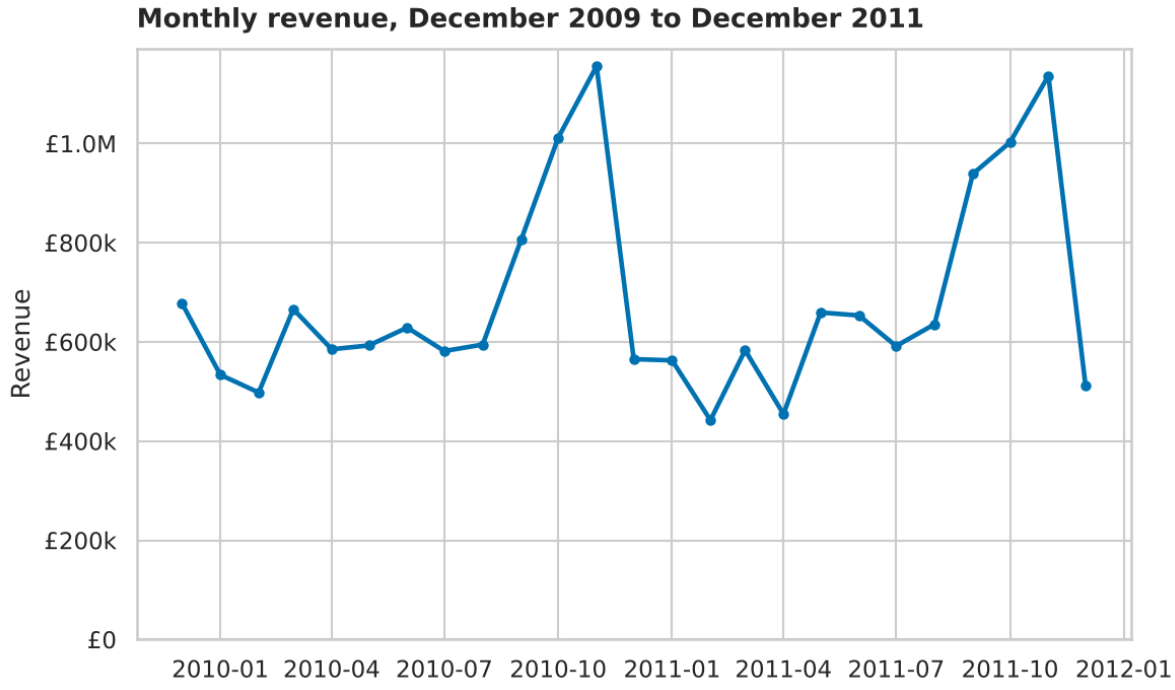


Figure 2. Monthly revenue, December 2009 to December 2011 (identified customers; first and last months partial).

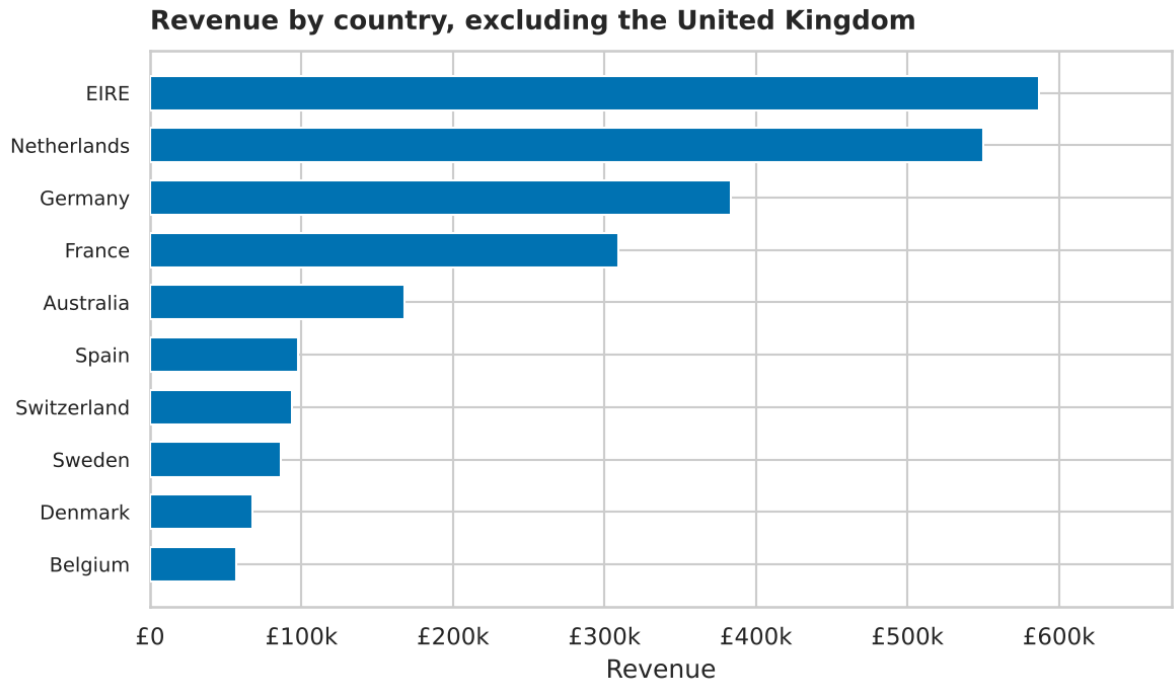


Figure 3. Revenue by country, excluding the dominant UK market (identified customers).

The most important pattern for segmentation is concentration. Ranking customers from highest to lowest spend and accumulating their revenue shows that the top 10% of customers already account for about 64% of revenue, and the top 20% for 77% (Figure 4). This is the practical case for

segmentation: a flat view of the customer base hides the fact that a minority of buyers carries the business, and a uniform marketing approach would treat them the same as one off bargain hunters. The product mix is consistent with a gift retailer, led by decorative homeware, seasonal items and novelty lines, with no single product dominating revenue.

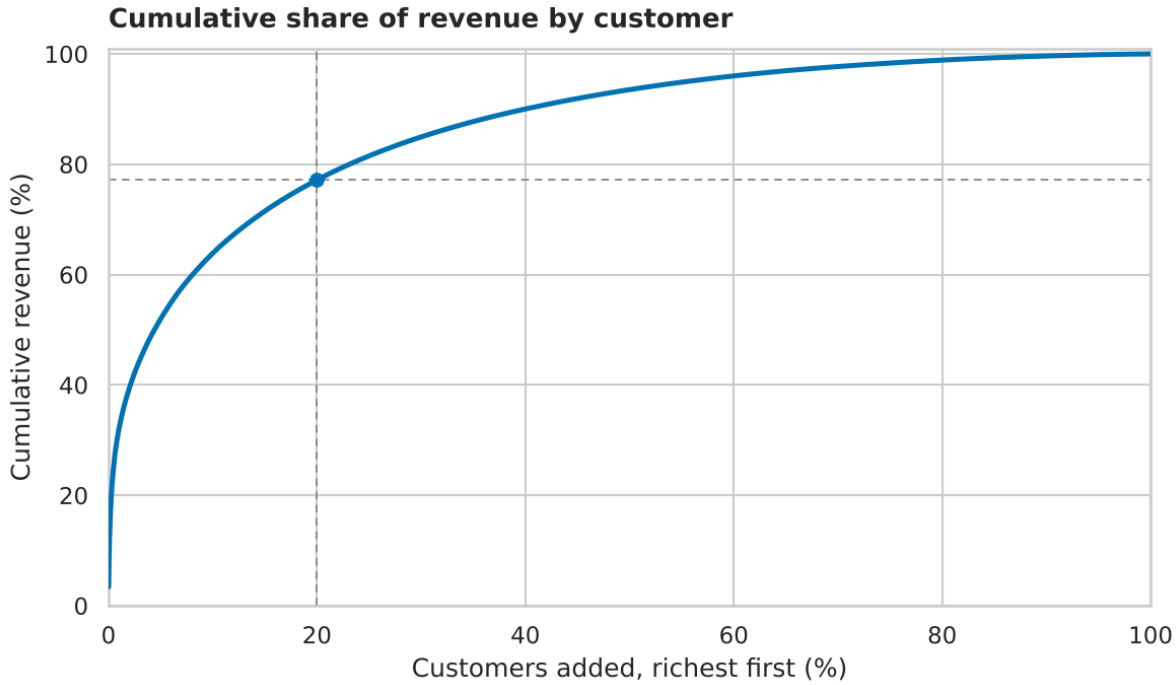


Figure 4. Cumulative share of revenue across the customer base, richest first.

4. Modelling Approach

We reduced each customer to three behavioural features. Recency is the number of days between the customer's last order and a fixed snapshot date, taken as the day after the final transaction (10 December 2011), so the measure is stable. Frequency is the number of distinct orders, and Monetary is total revenue. These features come from domain knowledge about what a retention team can influence, not from the algorithm, which is where the useful signal in a model like this usually originates.

Two preparation steps matter. All three features are heavily right skewed, with a small number of wholesale sized customers stretching the scale, so we applied a log transform to make the distributions roughly symmetric. We then standardized the features, because K-Means measures distance and the unscaled Monetary value, which runs into the thousands, would otherwise dominate Recency and Frequency and silently decide the clusters on its own. We also tested whether adding Average Order Value or customer Tenure improved the clustering. Both are derived from R, F and M, so they add little independent signal: Average Order Value lowered the silhouette (0.28 against 0.37 for RFM alone) and Tenure raised it only marginally (0.39), a difference inside the noise of a subsampled silhouette estimate. We kept the parsimonious, business legible RFM frame the brief calls for.

As a transparent reference point we first scored every customer from one to four on each of R, F and M by quartile, a rule based segmentation that needs no model. K-Means agrees with this baseline only loosely (the adjusted Rand index of 0.27 partly reflects that the quartile score has more levels than

there are clusters), and the clearest evidence is in the crosstab: about 79% of the commercially important At-Risk segment falls in the baseline's middle tier, so a plain quartile cut would bury these lapsing but valuable customers in the middle of the pack rather than isolating them. That is the structure K-Means adds. For the clustering itself we chose the number of clusters with the elbow method (Thorndike, 1953), plotting the within cluster sum of squares against k (Figure 5). The elbow is soft, and the silhouette score is in fact highest at k equal to two (0.44 against 0.37 at k equal to four), but a two way split only separates engaged from lapsed customers, which is too coarse to drive different actions. Following the principle that the business question should set the granularity, we chose k equal to four. The resulting silhouette of 0.37 indicates moderate, not sharp, separation, which is expected for behavioural data and is why we treat the clusters as a starting point for interpretation rather than an answer. The solution is stable: across ten random initializations the assignments agree almost perfectly (mean adjusted Rand index 0.99), and removing the top 1% highest spenders and re-clustering reproduces the same structure (0.91), so the segments are not an artifact of one initialization or a few outliers. The final model is a single scikit-learn pipeline (Pedregosa et al., 2011) chaining the log transform, the scaler and K-Means, so that the identical steps apply to any new customer. The analysis was implemented in Python with pandas (McKinney, 2010), scikit-learn and mlxtend.

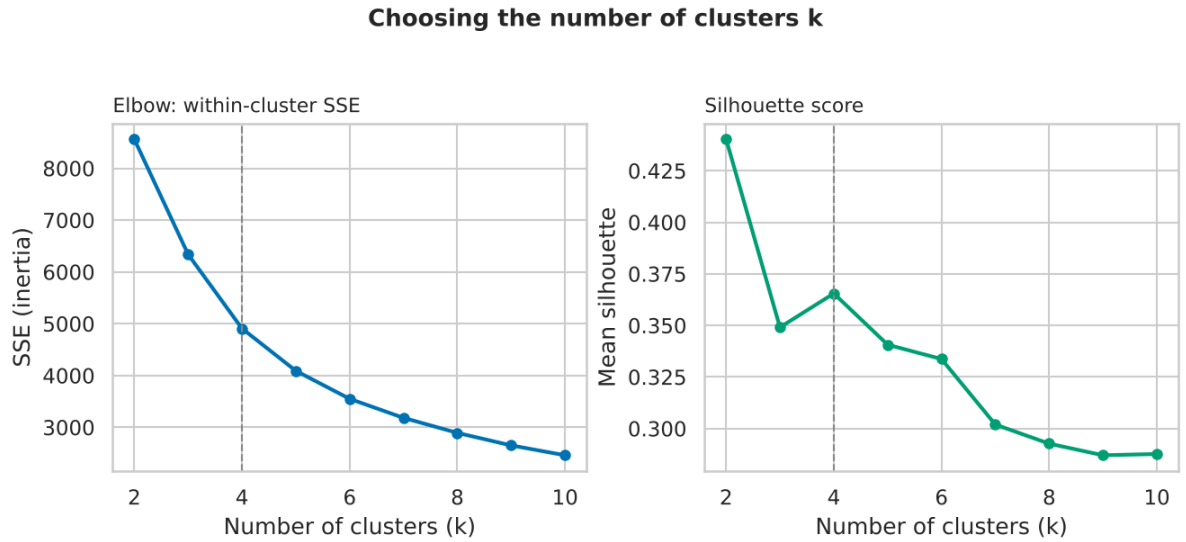


Figure 5. Elbow (within-cluster SSE) and silhouette score by number of clusters. Silhouette peaks at k = 2; k = 4 was chosen for actionable granularity.

5. Results and Discussion

The four segments are clearly visible on recency and frequency (Figure 6) and have distinct fingerprints across the features (Figure 8). We read each cluster from its own profile rather than labelling it by rule: one group is best on all three measures (Champions), one is worst on all three (Hibernating), and of the remaining two the lapsed, higher recency group is At-Risk while the recent, lower spending group is Loyal or Promising. Their profiles and the actions they imply are summarised in Table 1.

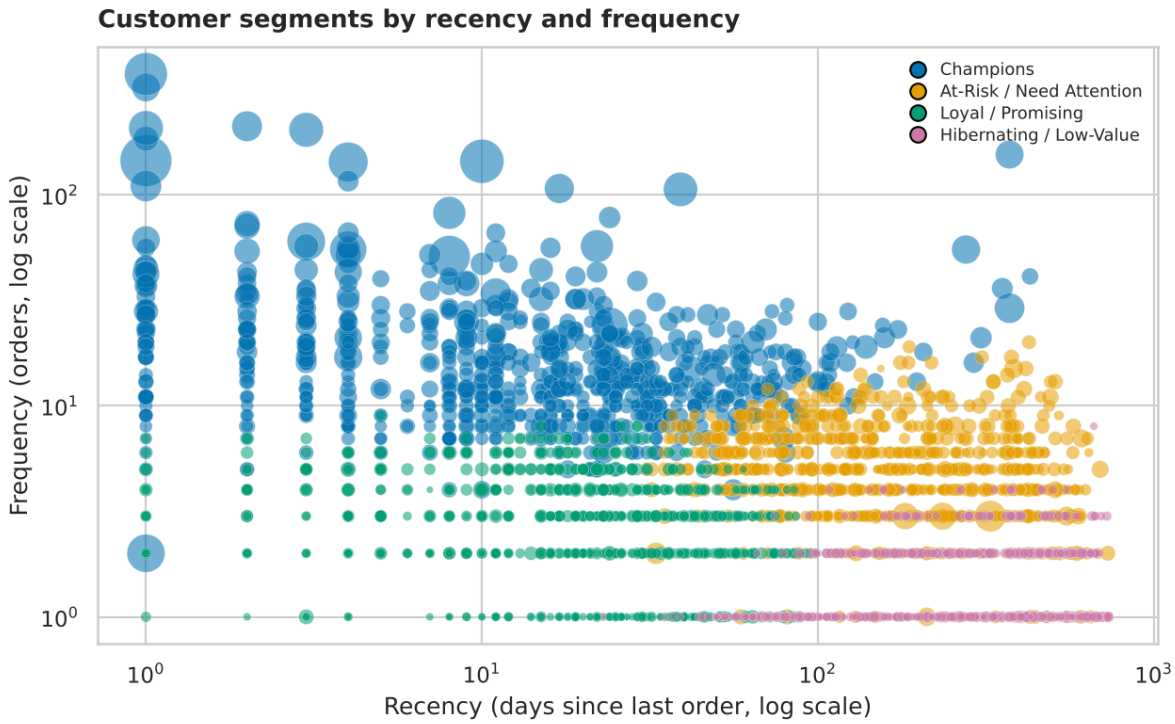


Figure 6. The four segments on recency and frequency; marker size is total spend (both axes log scaled).

Table 1. Segment profiles (medians) and share of customers and revenue.

Segment	Customers	Revenue	Median R (days)	Median F	Median M
Champions	20%	73%	16	13	£4,905
At-Risk / Need Attention	25%	17%	184	4	£1,455
Loyal / Promising	21%	6%	24	3	£720
Hibernating / Low-Value	34%	4%	401	1	£276

The headline is concentration. Champions are about one in five customers but almost three quarters of revenue: recent, frequent and high spending. They need protecting rather than discounting, through a loyalty tier, early access to new ranges and reliable service, since a discount here simply gives away margin on sales that would happen anyway. The At-Risk group is the most commercially interesting. These customers were valuable, with a median of four orders and £1,455 spent, but their last purchase was around six months ago, so they are drifting away while they are still worth winning back, and together they represent about £2.9 million in historic revenue. A targeted, time limited offer with a personal reminder is the right move here, and it is where retention spend earns the most. The Loyal or Promising group buys recently but in small amounts, so the opportunity is to grow basket size through bundles and cross-sell. The Hibernating group is a third of customers but one off, low value buyers who went quiet more than a year ago, and the rational response is to reach them only through the cheapest channels rather than spend heavily to reactivate them.

A subtle but useful point sits in the average order values. Champions and At-Risk customers spend

similar amounts per order, about £366 and £321, so what separates them is not basket size but how recently and how often they buy. That is encouraging for retention, because frequency and recency are exactly what a well timed win-back can move, whereas basket size is harder to shift.

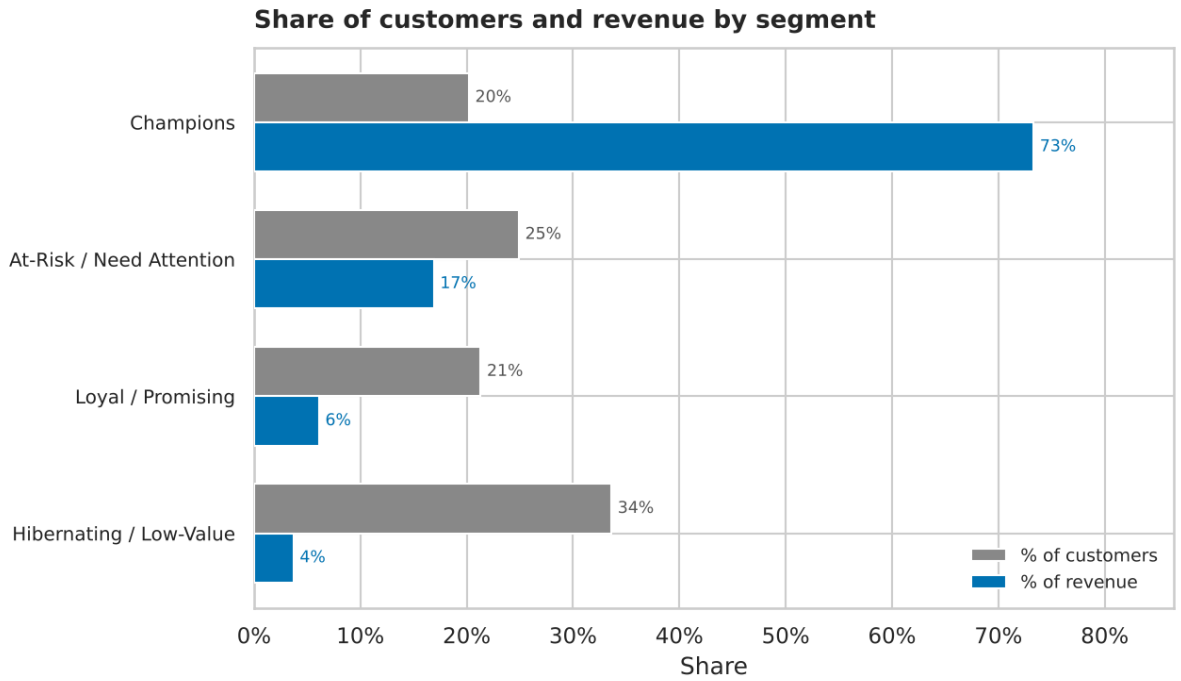


Figure 7. Share of customers against share of revenue, by segment.

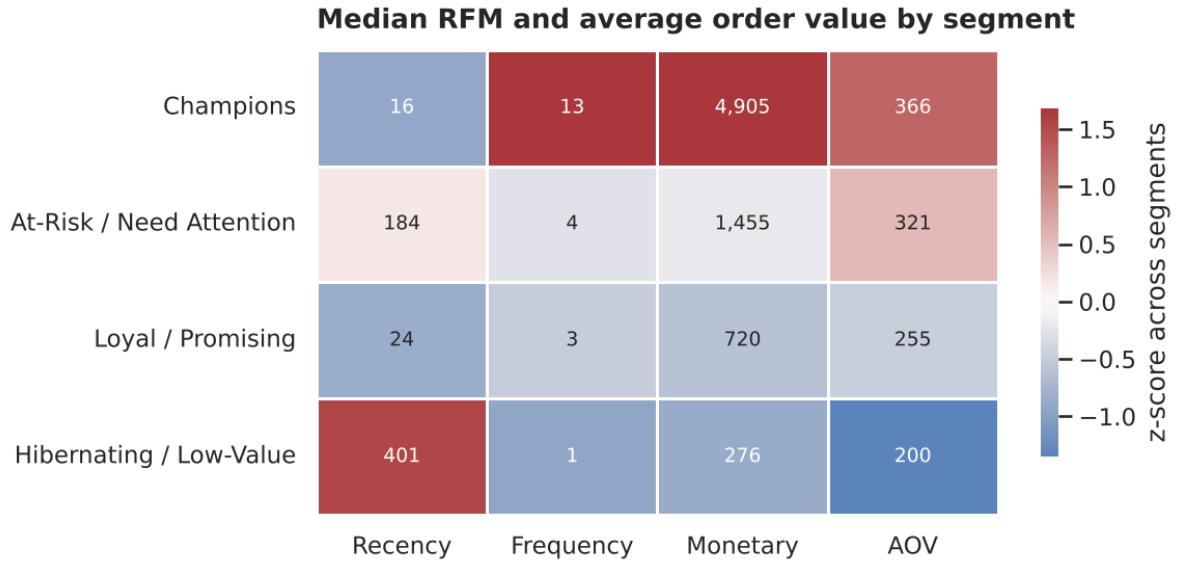


Figure 8. Segment fingerprints: median of each feature by segment, standardized across segments. For Recency a lower value is better (blue); for Frequency, Monetary and AOV a higher value is better (red).

One modelling choice we should defend is deleting cancellations rather than netting returns against purchases, which could in principle inflate a customer's Monetary value and mislabel a heavy returner. We checked it: 42% of customers returned something, but only 100 customers, about 1.7%, had returns worth more than a fifth of their gross spend, and only seven of those sit in the Champions segment. Netting returns would therefore not meaningfully change the segments, which justifies the simpler rule.

Because the data is transactional we also ran a market basket analysis on UK orders, ranking rules by lift rather than confidence, since a high confidence rule can be an artifact of a generally popular item while lift measures genuine association (Agrawal & Srikant, 1994), using the Apriori algorithm as implemented in mlxtend (Raschka, 2018). Lift is symmetric, so we removed mirror rules and kept one row per pair. The strongest associations are variant completions within a line, such as the green and pink Regency teacups at a lift of 27 and the Scandinavian wooden Christmas decorations, which are ready made prompts for completing a set. Lowering the support threshold surfaces genuinely cross-category rules as well, for example a decorative building block word bought with the white hanging heart t-light holder, and a bathroom metal sign with a kitchen one, which point to home decor cross-sell rather than mere set completion, though at more modest lift than the within-line pairs.

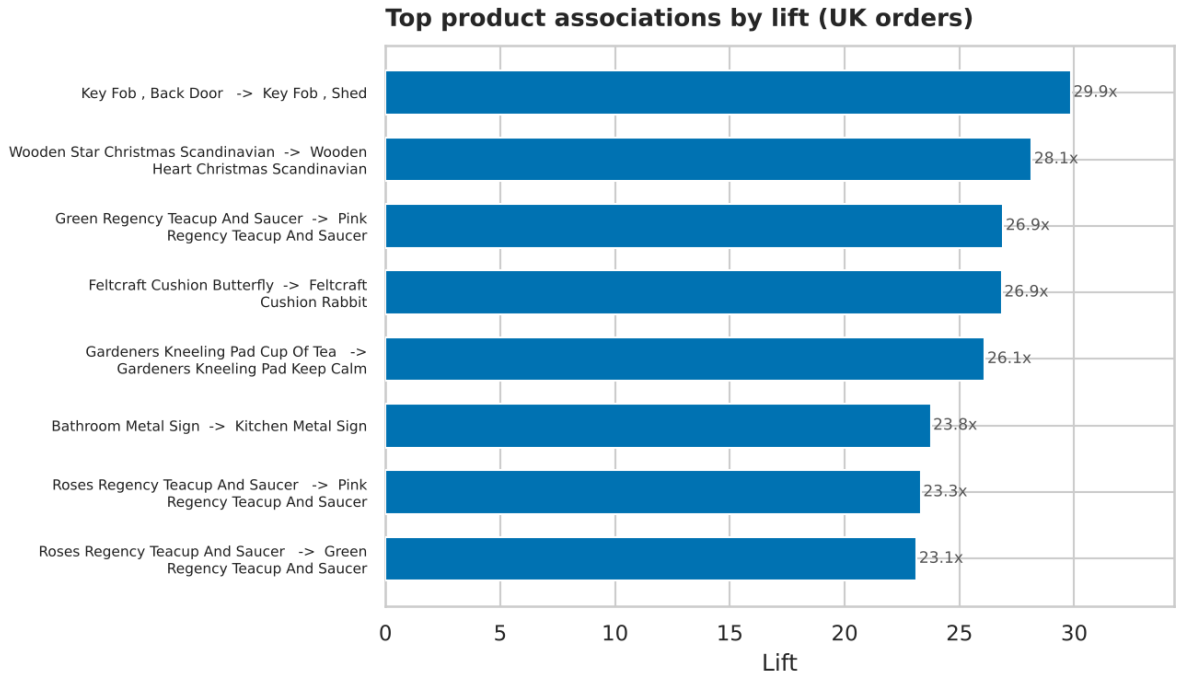


Figure 9. Product pairs bought together, ranked by lift, among UK orders (mirror rules removed).

Correlation is not causation. The segments describe behaviour, they do not explain it. Contract length, product category and customer type are entangled, so we can say that Champions buy often and spend heavily, but not that any single feature causes loyalty. The model is a map of who to treat differently, not a theory of why they behave as they do.

On fairness, the model uses only behavioural fields and no demographic attributes, so it does not segment on protected characteristics directly. It can still reflect them indirectly, for example if wholesale buyers cluster by region, so any intervention built on these segments should be monitored for unintended skew before it is scaled.

6. Limitations: What the Data Cannot Tell Us

Several limitations bound what this analysis can claim. First, roughly a fifth of transactions have no customer identifier and are excluded from the segmentation; they account for about 13% of valid revenue, so the customer-level view is not the whole business. Second, the data has no cost or margin, so value is measured as revenue rather than profit, and a high-revenue segment is not automatically high-margin. Third, there is no browsing, marketing-spend or return-reason data, so the model

describes what customers did, not why, and cannot attribute sales to campaigns. Fourth, the business is heavily UK-weighted, about 84% of revenue, so the patterns should not be generalised to other markets, and the data covers a single giftware retailer between 2009 and 2011, so it is both specific and dated. Finally, the clusters are a decision aid, not ground truth: the silhouette of 0.37 indicates moderate separation, and although the four segments are stable across initializations and to outlier removal, they should steer experiments rather than replace them.

7. Conclusion and Recommendations

A disciplined clean reduced 1.07 million messy rows to 776,577 reliable transactions, and RFM with K-Means turned 5,852 customers into four segments that each demand a different response. The single most important finding is the concentration of value: roughly 20% of customers generate about 73% of revenue, so retention effort and budget should be weighted heavily toward Champions, with the clearest growth opportunity in winning back the At-Risk group, which holds about £2.9 million in historic revenue, before it lapses. The market basket rules give the merchandising team specific pairs to bundle today.

We recommend four concrete steps. First, stand up a Champions retention programme measured on repeat rate rather than discount depth. Second, launch a win-back campaign aimed at the At-Risk segment with a defined inactivity trigger, and measure its lift against a holdout group. Third, use the association rules to drive bundle and cross-sell placements, starting with the set completion pairs and the cross-category home decor rule. Fourth, cap reactivation spend on the Hibernating segment. Beyond these actions, the most valuable next step for the analysis is to bring in data the current dataset lacks, above all product cost so that value can be measured as profit rather than revenue, and campaign and browsing data so that the business can begin to explain behaviour and not only describe it.

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